

Israel Vallejo Muñoz

Marine Biologist (MSc) · Environmental & Coastal Data Analyst

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PROFESSIONAL SUMMARY

Coastal data analyst who quantified about 330,000 ha of mangrove loss and 582 Mt CO₂e across nine Southeast Asian countries using Global Mangrove Watch. MSc in Marine Biology and Aquaculture. I combine Sentinel-1/2 imagery, Google Earth Engine, QGIS and R to turn coastal wetland change into evidence for conservation decisions. Experience in Southeast Asia and Mexico, covering mangroves, coastal aquaculture and marine protected areas. Based in Mexico, available to work remotely across time zones.

CORE COMPETENCIES

Coastal wetlands: mangrove mapping and loss, blue carbon (IPCC Tier 1), Mann-Kendall and Sen's slope, coastal aquaculture suitability

Remote sensing and GIS: Google Earth Engine, Sentinel-1 SAR, Sentinel-2, QGIS, R (terra, sf, tidyverse), Global Mangrove Watch

Marine biology and fisheries: elasmobranch ecology and bycatch, shark and ray field sampling (tagging, biopsy, morphometrics), reproductive biology, fish identification, CPUE standardisation, stock assessment (Schaefer, Fox), MSY

Protected areas: Global Fishing Watch, WDPA, fishing-effort analysis in marine protected areas

PROFESSIONAL EXPERIENCE

Fisheries Data Analyst (External Collaboration)

Jan 2026 – Present

Sea Shepherd Global · Remote

- Analysed fisheries and compliance data across the Mediterranean Sea and the African coast, giving the international team evidence for its campaign strategy.
- Turned standardised fishing-effort analyses into reports for international scientific teams.
- Built spatial visualisations of fishing effort and compliance metrics in QGIS and Google Earth Engine.

Environmental & Fisheries Data Analyst

Jan 2025 – Present

Independent Consultant · Remote

- Estimated about 330,000 ha of net mangrove loss and 582 Mt CO₂e across nine Southeast Asian countries (1996-2020) using Global Mangrove Watch v3.0, Mann-Kendall testing and country-specific emission factors, with significant trends in four of nine countries.
- Compared mangrove loss and carbon stock reduction in the Mekong Delta (-9.0%) and the Mexican Pacific coast (-7.6%) using GMW v3.0 and IPCC Tier 1 coefficients.
- Integrated Sentinel-1 SAR with Sentinel-2 optical imagery in Google Earth Engine to detect coastal vegetation change in Ao Phang Nga Bay, Thailand (2017-2024), finding 84.7% stable area, 7.5% gain and 7.8% loss, with statistically significant change across NDVI, EVI and SAR indices.
- Built an AHP-WLC suitability model for offshore cage shrimp (*Penaeus vannamei*) aquaculture in the Gulf of Thailand using sea-surface temperature, bathymetry, chlorophyll-a and distance to ports.
- Mapped vegetation loss in Central Borneo (2018-2024) with Sentinel-2 NDVI differencing, cross-checked across Google Earth Engine, R and QGIS.
- Analysed the overlap between fishing effort and marine protected areas in the Mexican EEZ (21,565 hours, 3,663 grid cells, 2024 GFW data), finding 18.1% of effort inside MPAs.
- Ongoing: analysing fishing activity in Thai marine protected areas with Global Fishing Watch data, using buffer zones and Mann-Kendall trend tests.
- Built fisheries stock assessment models (Schaefer and Fox), estimating MSY at 56,300 t for Eastern Pacific bigeye tuna.
- Developing an offline-capable digital tool for small-scale fisheries and aquaculture management, covering catch logging, regulatory compliance reporting and spatial data capture.

Research Intern & Teaching Assistant

Feb – Oct 2024

Università degli Studi di Napoli Federico II · Naples, Italy

- Processed tissue samples (spectrophotometric, chromatographic, histological techniques) assessing reproductive physiology in aquaculture and wild fish; six months as teaching assistant in marine biology laboratory instruction.
- As a member of the laboratory, coordinated meetings and discussion materials between the research group and partner laboratories and institutions.

Social Service, Fish Identification & Curation

Jan – Nov 2019

Colección de Zoología, Facultad de Ciencias, UNAM · Mexico City

- Identified and taxonomically verified freshwater and marine fish specimens for the departmental zoological collection.
- Supported specimen curation, cataloguing and long-term maintenance of the collection.

Research Assistant, Shark Ecology

Jul – Aug 2018

Shark Research and Conservation Program · Miami, USA

- Captured, tagged and sampled 15+ large coastal sharks across 6–7 species, collecting blood, biopsy and morphometric data for ecological research.

Research Assistant, Ichthyology

Apr 2016 – Jan 2018

National Fish Collection & ICMYL, UNAM · Mexico City

- Processed and identified freshwater and marine fish specimens, including multiple tilapia species, extracted otoliths for age and growth studies, applied bone clearing and staining (diaphanization) techniques, and contributed to curation and maintenance of the national ichthyological collection.

EDUCATION**MSc in Marine Biology and Aquaculture**

Oct 2024

*Università degli Studi di Napoli Federico II · Naples, Italy*Thesis: Recent record of *Coris julis* in Mellah Lagoon: role of anatomical studies in a Changing Climate.**BSc in Biology**

Jan 2022

*Universidad Nacional Autónoma de México (UNAM) · Mexico City*Thesis: Reproductive biology of the deep-sea shark *Etmopterus schultzi* in the Gulf of Mexico.

ADDITIONAL SKILLS & CERTIFICATIONS

DIVING	NADD Advanced Open Water Diver, PADI Open Water Diver
LANGUAGES	Spanish (Native/C2), English (Proficient/C1, TOEFL iBT 100), Italian (Intermediate/B2)
OTHER CERTIFICATIONS	GIS Applied to Natural Resources & AI Integration; GHG Protocol: Foundations; CSRD & ESRS Reporting; FAO eLearning Academy, Management Procedures for Sustainable Tuna Fisheries (5-course series)